Systems, vectors & linear mappings

Printable study reference. The supplied documents do not confirm examination permission.

01 Read the system, then reduce

A is m × n: m equations, n unknowns. In [A | b], the final column is data, not an extra unknown.

Legal row operations: interchange two rows; multiply a row by c ≠ 0; add c times another retained row. Apply the operation to the entire row, including b.

Forward: move left to right, place a nonzero pivot, clear below; skip an unavailable column. **Backward:** normalize pivots to 1 and clear above.

02 REF, RREF & pivot locations

REF: zero rows last; leading entries move strictly right; zeros below every leading entry. Leading entries need not be 1.

RREF: REF + each leading entry is 1 + each pivot 1 is alone in its column. RREF is unique; ordinary REF is not.

Pivot columns: use the original column indices corresponding to leading 1s in RREF. A numerical pivot during elimination may change.

03 Read the answer after elimination

No solution: an echelon row [0 ... 0 | c], c ≠ 0. Equivalently, the appended column is a pivot column.

Consistent + no free variables: one solution. **Consistent + a free variable:** infinitely many. A zero row [0 ... 0 | 0] is harmless.

If rank(A) = r, the number of free variables is n - r. Assign a different real parameter to each free variable.

Template: $x = p + s_1v_1 + ... + s_k v_k$, where $Ap = b$ and $Av_j = 0$. Homogeneous: b = 0, so p may be 0.

04 A complete parameter example

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$x_2 = s$, $x_3 = t$; $x_1 = 4 - 2s + t$. Thus $x = (4,0,0) + s(-2,1,0) + t(1,0,1)$, $s,t \in \mathbb{R}$.

Check: (4 - 2s + t) + 2s - t = 4. The redundant second equation adds no restriction.

05 Vectors, combinations & span

Add matching coordinates; a scalar multiplies every coordinate. cu + dv is a linear combination. Zero weights are allowed.

Ax = b means $x_1a_1 + ... + x_n a_n = b$, with a_j the columns of A. Solve [A | b] for the weights.

$b \in$ Span$\{a_1,...,a_n\}$ ⇔ Ax = b is consistent. One nonzero direction spans a line; two independent directions span a plane. Every span contains 0.

06 Independence = no redundant direction

Test: solve Ac = 0 with the vectors as columns. Only c = 0 ⇒ independent. A solution with some $c_j \neq 0$ ⇒ dependent.

Dependence relation: $c_1v_1 + ... + c_p v_p = 0$ with weights not all zero. Rearrange a nonzero-weight term to exhibit redundancy.

Any collection containing 0 is dependent. Two vectors are dependent iff one is a scalar multiple of the other. More than n vectors in $\mathbb{R}^n$ are dependent; the converse is false.

07 From a rule to a standard matrix

Linearity: T(u + v) = T(u) + T(v) and T(cu) = cT(u), for all allowed u,v,c. T(0) = 0 is necessary, not sufficient.

$A = [T(e_1) ... T(e_n)]$. Domain $\mathbb{R}^n$ counts columns; codomain $\mathbb{R}^m$ counts rows. Then T(x) = Ax.

Composition: first S, then T gives $A_T A_S$. The rightmost action happens first.

08 Keep existence & uniqueness separate

For A of size m × n and T: $\mathbb{R}^n \to \mathbb{R}^m$:

Onto ⇔ columns span $\mathbb{R}^m$ ⇔ pivot in every row ⇔ Ax = b solvable for every b in the codomain.

One-to-one ⇔ columns independent ⇔ pivot in every column ⇔ Ax = 0 has only the trivial solution.

A 3 × 5 matrix with 3 pivots is onto $\mathbb{R}^3$, with 2 free variables. Every b is attainable, but no solution is unique.

Matrix algebra, inverses & methods

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09 Dimensions before arithmetic

Addition: same dimensions; add entries. **AB:** $(m \times n)(n \times p)$ gives $m \times p$. Inner dimensions must match.

Entry (i,j): row i of A dotted with column j of B: $(AB)_{ij} = \Sigma$ from k=1 to n of $a_{ik} b_{kj}$.

Example: [2,3] paired with (4,1) gives $2 \cdot 4 + 3 \cdot 1 = 11$. Column j of AB is A b_j.

10 Algebra rules & common traps

Valid: $A(BC) = (AB)C$; $A(B + C) = AB + AC$; $(A + B)C = AC + BC$; $AI = IA = A$ (matching sizes).

Generally false: $AB = BA$; $AB = 0 \Rightarrow A = 0$ or $B = 0$; $AB = AC \Rightarrow B = C$. Cancellation needs an invertible factor on the correct side.

Transpose: $(A^T)^T = A$; $(A + B)^T = A^T + B^T$; $(cA)^T = cA^T$; $(AB)^T = B^T A^T$.

11 Inverse: a two-sided undo operation

Only square matrices have the ordinary inverse. $A^{-1}A = AA^{-1} = I$. For A = [a,b;c,d], the inverse exists iff $ad - bc \neq 0$.

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Gauss-Jordan: reduce [A | I] to [I | A^{-1}]. Carry every operation through both blocks. If A cannot reduce to I, it is singular.

Solve: $x = A^{-1}b$. **Selected column j:** solve $Ax = e_j$; there is no need to compute all of A^{-1}.

12 Inverse identities (invertible factors)

$(AB)^{-1} = B^{-1}A^{-1}$; $(A^T)^{-1} = (A^{-1})^T$; $(A^{-1})^{-1} = A$.

Check an inverse: multiply with the original A in both orders. Do not take reciprocals of entries.

13 Plane transformations: x ↦ Ax

Horizontal-axis reflection	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Vertical-axis reflection	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
Reflection in $x_2 = x_1$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Horizontal projection	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
Horizontal shear (factor k)	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$
Scale both coordinates by k	$\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$
Counterclockwise rotation θ	$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Clockwise rotation uses a negative θ. A positive scale $k > 1$ stretches; $0 < k < 1$ contracts. Inverse shear replaces k by -k.

14 Choose a method, then verify

System / span / range: form [A | b] → reduce → check consistency → parameterize if needed → verify $Ap = b$ and $Av_j = 0$.

Independence / one-to-one: solve $Ac = 0$; decide whether any nonzero weight vector exists.

Onto / every b: check for a pivot in every coefficient row, not merely every row of [A | b].

Proof: state assumptions → use the definition → justify each implication → return to the claim. For "iff", prove both directions.

True/false: a true universal claim needs a general reason; a false one needs a valid counterexample. One successful example never proves "for all".

15 Before writing the final line

Check dimensions, signs, free parameters, assumptions, and factor order. State "no solution", "one solution", or the full parameterized set as the question requires.

Source scope: sections 1.1-1.5, 1.7-1.9, 2.1-2.2. No 1.6. The lecture inverse typo on physical page 98 is explained in the reader.